DIPLOMA PROGRAMME PRACTICE / HL / SET A

Economics / Paper 1

Worked mark scheme · 75 minutes · 25 marks

Choose one of the three questions and answer both parts. No calculator is permitted. Use real-world examples in part (b) and fully labelled diagrams where relevant. Write on separate answer paper.

Judge extended responses as a whole using the markband for that part. Award valid alternative analysis and diagrams. In linked calculations, credit correct use of an earlier result. The suggested judgements are examples.

HOW TO MARK THIS PAPER

Use the 10-mark scale for each explanation and the 15-mark scale for the evaluation. Only the two parts of the selected essay contribute to the score.

10-MARK SCALE

0: no relevant economics. 1–2: isolated relevant ideas. 3–4: partial explanation with limited links. 5–6: a broadly correct mechanism, with gaps in development or representation. 7–8: developed analysis using appropriate terms and an effective diagram or other representation where relevant. 9–10: consistently precise, fully connected reasoning directed at this question, with the representation interpreted. Accept equivalent valid analysis.

15-MARK SCALE

0: no relevant economics. 1–3: assertions or isolated concepts. 4–6: some relevant explanation, limited application. 7–9: coherent analysis with some application and consideration of consequences. 10–12: developed, applied analysis weighing competing considerations with a supported view. 13–15: precise theory and relevant diagrams where useful, evidence integrated throughout, and a balanced judgement that explains conditions or relative importance. Other defensible conclusions can receive full credit.

QUESTIONS IN THIS SCHEME

1 Regulating a network monopoly [25]

2 Stabilising a fall in expenditure [25]

3 Protection and the distribution of surplus [25]

QUESTION 1 / 25 MARKS

Regulating a network monopoly

Background reading. Ofwat, Price reviews

PART a / 10 MARKS

Explain why the profit-maximising output of a single-price monopolist can differ from the allocatively efficient output.

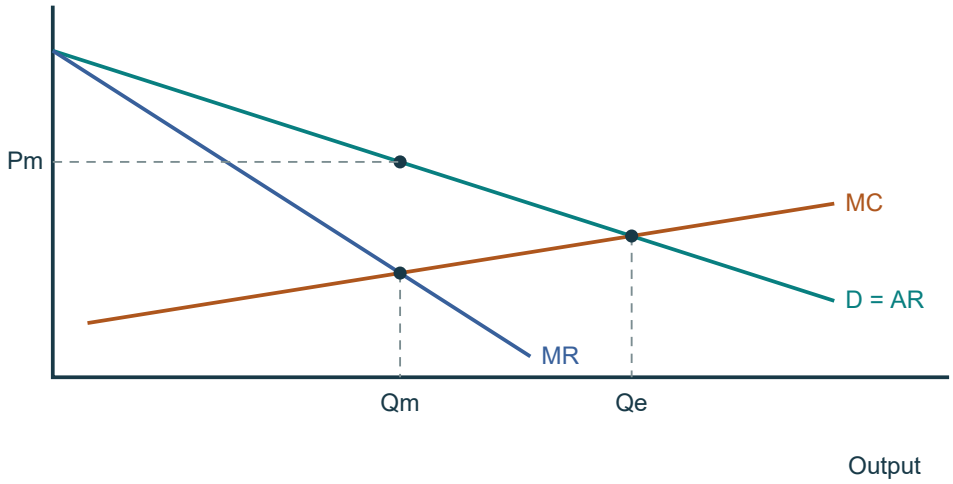
Analysis. With downward-sloping demand the extra revenue from another unit is below its price because selling more normally requires a lower price on earlier units. A profit-seeking firm expands to MR = MC on the relevant rising marginal-cost range, then reads its price from demand. Without externalities the efficient output satisfies price = MC.

Diagram. On price/revenue/cost and output axes draw AR = demand, a steeper MR below it and MC. Mark Qm at MR = MC, Pm on demand and Qe at demand = MC. For Qm to Qe, willingness to pay exceeds the opportunity cost of producing the unit. Monopoly profit itself is not the welfare loss.

Apply the 10-mark scale on the first page to this response as a whole.

WORKED REPRESENTATION / a

Price, revenue and cost



PART b / 15 MARKS

Using real-world examples, evaluate price regulation as a means of improving efficiency in a natural monopoly.

Application. Water networks have substantial sunk distribution costs and expensive duplication. Ofwat's periodic price reviews in England and Wales show how a regulator can link allowed charges to expenditure and service obligations. Lowering a monopoly price may expand consumption and reduce the output restriction.

Evaluation. Where average cost falls over the relevant range, setting price at marginal cost may require a subsidy to cover total cost. Average-cost pricing trades some allocative efficiency for viability. A multi-year cap can reward cost savings but a poorly calibrated cap may encourage service cuts, postpone maintenance or fund excessive returns.

Judgement. Choose the benchmark in light of cost information, service quality and investment needs. A cap paired with enforceable quality standards and periodic review is more credible for a water network than a low fixed price irrespective of costs. Credit a justified public ownership or regulation comparison.

Apply the 15-mark scale on the first page to this response as a whole.

QUESTION 2 / 25 MARKS

Stabilising a fall in expenditure

Background reading. European Commission JRC, Short-time work and discretionary policy in Germany

PART a / 10 MARKS

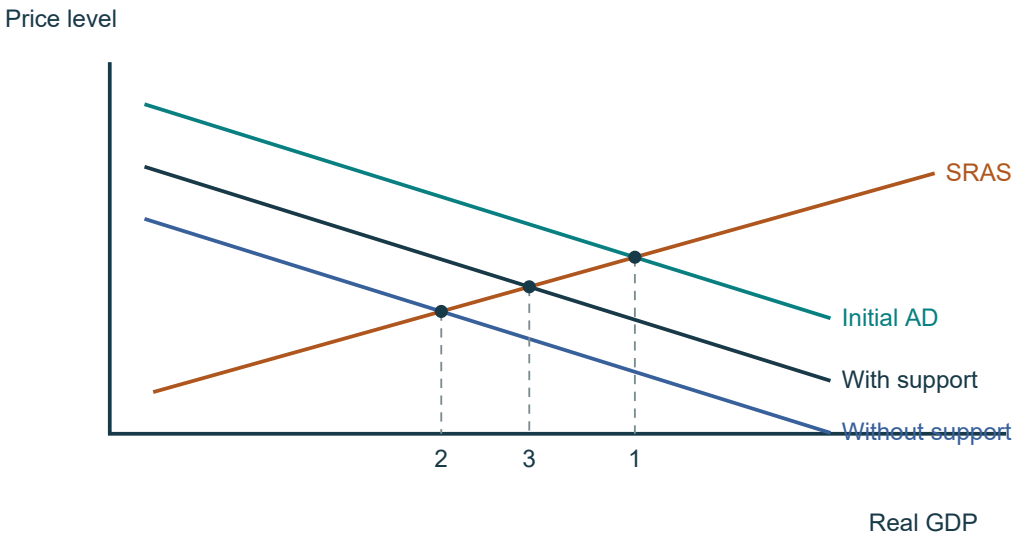
Explain how progressive income taxes and unemployment benefits can moderate a fall in aggregate demand without a new budget decision.

Analysis. As earned income falls, tax payments fall; in a progressive system the effective tax burden may also decline. Eligible unemployed workers receive transfers. Relative to unchanged net taxes, households lose less disposable income. Consumption falls less and subsequent rounds of lost income are smaller.

Diagram. Contrast AD after a negative demand shock without stabilisers with AD after the same shock operating through taxes and benefits. The latter is still left of the starting AD but farther right than the no-stabiliser counterfactual. SRAS is initially unchanged.

Apply the 10-mark scale on the first page to this response as a whole.

WORKED REPRESENTATION / a



PART b / 15 MARKS

Using real-world examples, discuss whether discretionary fiscal support should be preferred to automatic stabilisers in a deep recession.

Application. Germany's short-time work scheme helped maintain job matches during the pandemic. Distinguish payments under established eligibility rules from discretionary decisions to expand eligibility or duration. An announced wage subsidy or public investment can target a shock beyond the reach of ordinary unemployment benefits.

Evaluation. Automatic responses are quick and reverse as incomes recover, but informal workers or a sector suffering a mandatory closure may receive little support. Discretionary programmes can preserve relationships with employers; administrative delays, debt costs and supporting jobs that will not return can reduce the benefit.

Judgement. When demand has collapsed and spare capacity is extensive, a prompt, temporary targeted addition to existing stabilisers is defensible. If the activity cannot resume, training and mobility support may dominate an open-ended wage subsidy. Compare the size, duration and financing of the response.

Apply the 15-mark scale on the first page to this response as a whole.

QUESTION 3 / 25 MARKS

Protection and the distribution of surplus

Background reading. WTO, Members discuss US steel and aluminium measures (2018)

PART a / 10 MARKS

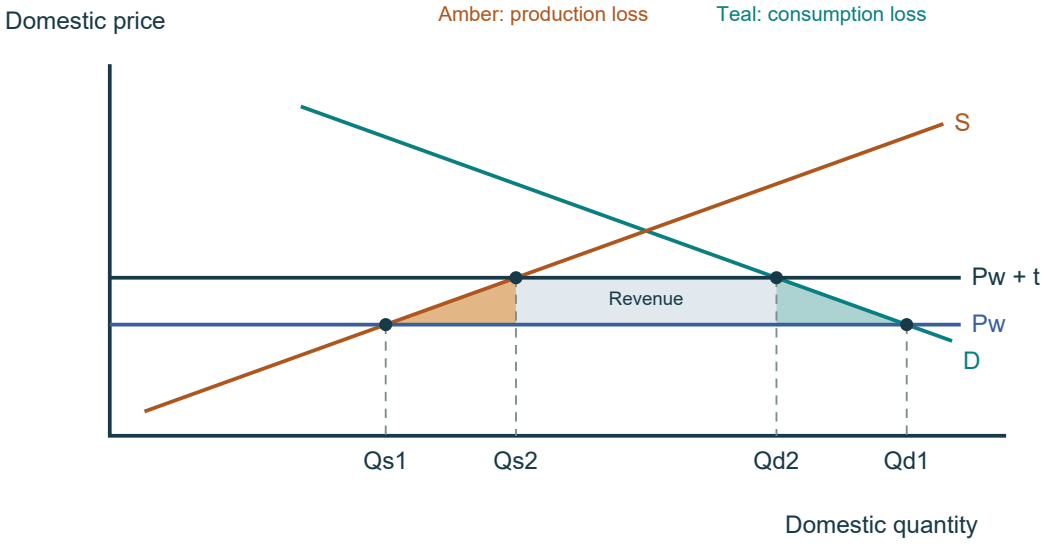
Using a diagram, explain the changes in domestic producer surplus and government revenue when a small importing country imposes a non-prohibitive specific tariff.

Analysis. World supply is perfectly elastic at Pw. With full pass-through the price becomes Pw + t. Domestic producers expand from Qs1 to Qs2 and receive more on the units they sell, increasing producer surplus by the area above supply between the two price lines. Domestic consumption falls from Qd1 to Qd2.

Diagram. Identify imports after the tariff as Qd2 − Qs2. Revenue is t times these remaining imports, not t times original import volume or all domestic sales. Locate the production and consumption deadweight-loss triangles separately from the transfers to producers and the government.

Apply the 10-mark scale on the first page to this response as a whole.

WORKED REPRESENTATION / a



PART b / 15 MARKS

Using real-world examples, evaluate the case for a tariff intended to protect employment in a strategic industry.

Application. The United States imposed additional steel tariffs in 2018. Domestic steel mills can benefit from higher prices and may retain capacity. Steel users such as equipment makers face higher input costs, so jobs gained in the protected industry are an incomplete measure of national employment.

Evaluation. Capacity and security benefits must be weighed against the cost per job, retaliation and substitution toward imported finished products. A small-country welfare diagram captures efficiency losses but leaves strategic benefits outside the model. A training subsidy targets displaced workers, while strategic procurement could target minimum capacity.

Judgement. A narrowly targeted temporary tariff is easier to defend when the security benefit is evidenced, domestic capacity can respond and support has an exit test. An indefinite broad tariff is harder to justify if downstream losses and trading-partner responses outweigh that benefit.

Apply the 15-mark scale on the first page to this response as a whole.